IP Security

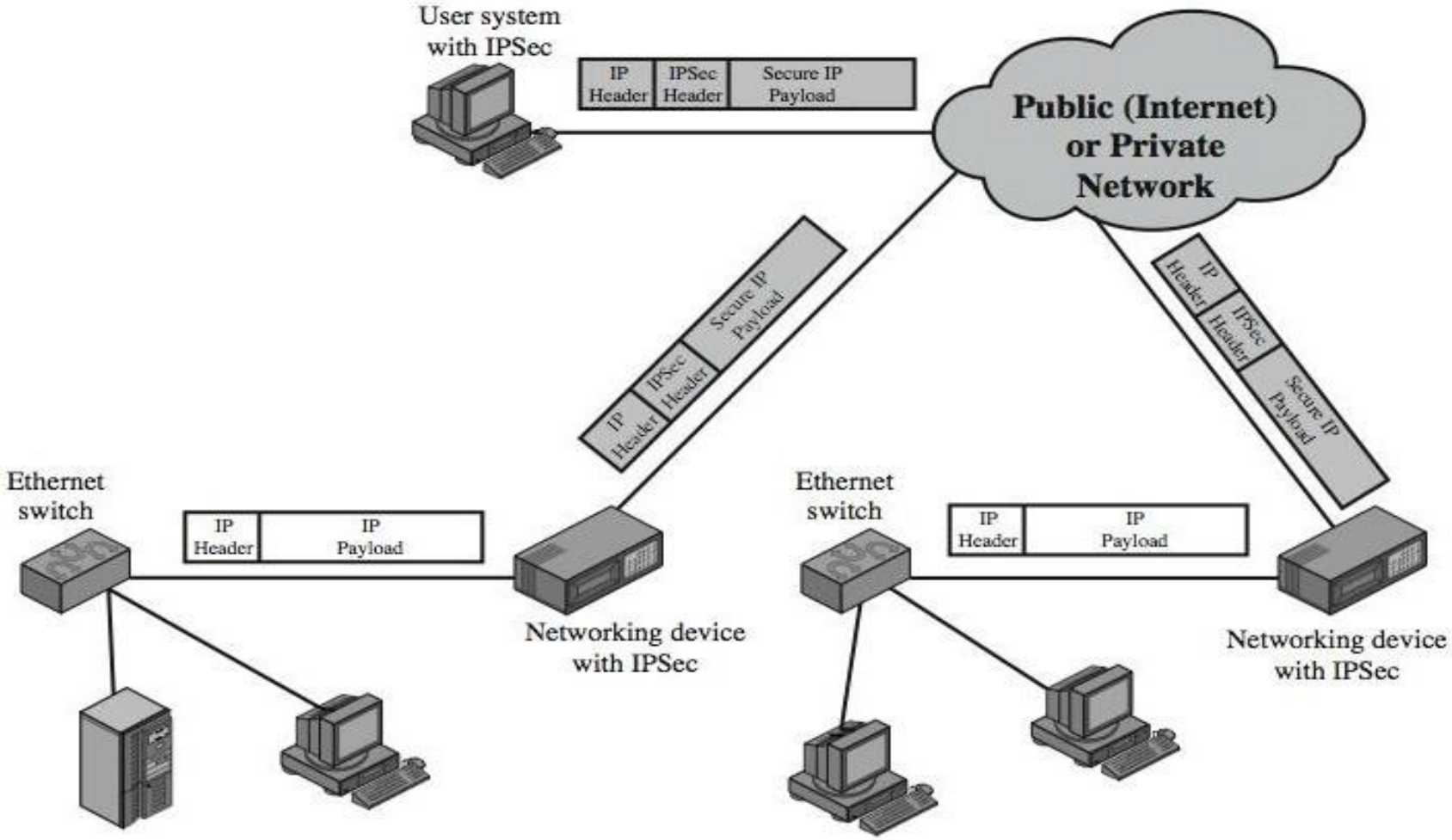
Overview

- Introduction to IP Security
- IPSec Services
- IP Security Policy – IPSec Architecture
  - Security Association Database
  - Security Policy Database
  - IP Traffic Processing
- Encapsulating Security Payload
  - ESP Packet Format
  - Encryption and Authentication
  - Padding
  - Anti-Replay Service
  - Transport and Tunnel Mode
  - Protocol Architecture
- Summary

Introduction to IP Security

- have a range of application specific security mechanisms
  - eg. S/MIME, PGP, Kerberos, SSL/HTTPS
- however there are security concerns that cut across protocol layers
- would like security implemented by the network for all applications
- general IP Security mechanisms provides
  - authentication
  - confidentiality
  - key management
- applicable to use over LANs, across public & private WANs, & for the Internet
- need identified in 1994 report
  - need authentication, encryption in IPv4 & IPv6

IP Security Uses



Benefits of IPSec

- in a firewall/router provides strong security to all traffic crossing the perimeter
- in a firewall/router is resistant to bypass
- is below transport layer, hence transparent to applications
- can be transparent to end users
- can provide security for individual users
- secures routing architecture

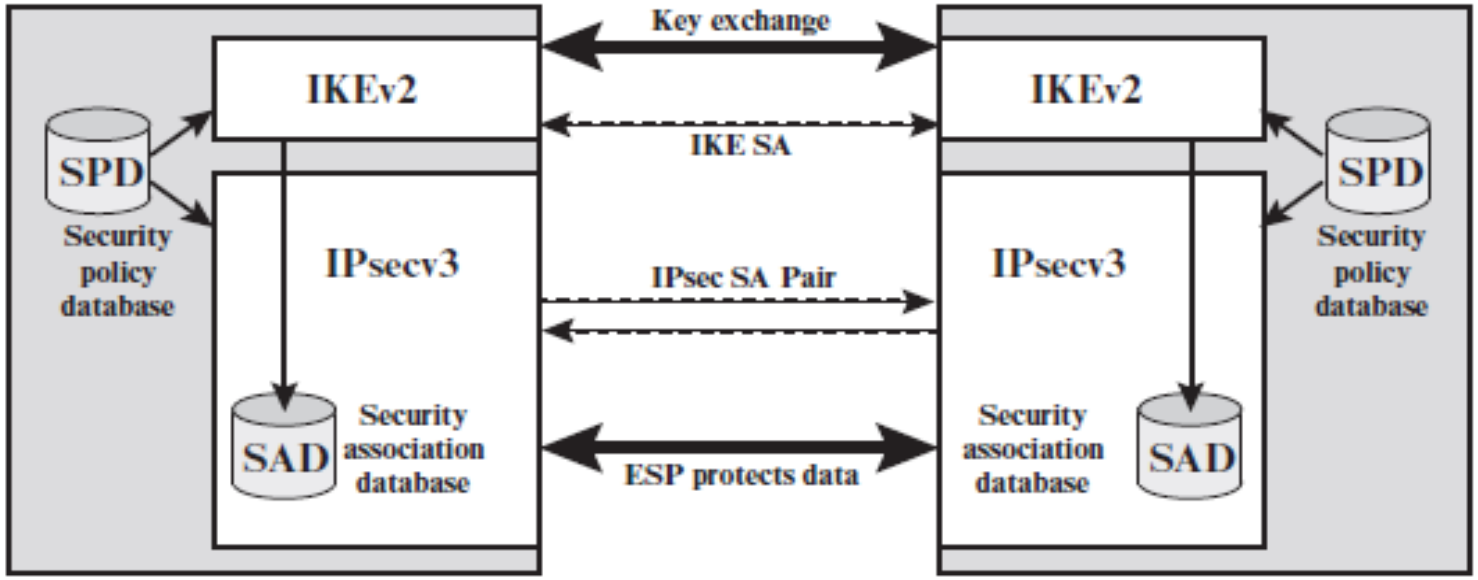
IPSec Services

- Access control
- Connectionless integrity
- Data origin authentication
- Rejection of replayed packets
  - a form of partial sequence integrity
- Confidentiality (encryption)
- Limited traffic flow confidentiality

IP Security Policy

- Fundamental to the operation of IPsec is the concept of a security policy applied to each IP packet that transits from a source to a destination.
- IPsec policy is determined primarily by the interaction of two databases, the **security association database (SAD)** and the **security policy database (SPD)**.

IPSec Architecture



Security Associations

- a one-way relationship between sender & receiver that affords security for traffic flow
- defined by 3 parameters:

Security Parameters Index (SPI)

✓ A 32-bit unsigned integer assigned to this SA and having local significance only.

✓ The SPI is carried in AH and ESP headers to enable the receiving system to select the SA under which a received packet will be processed.

IP Destination Address

✓ This is the address of the destination endpoint of the SA, which may be an end-user system or a network system such as a firewall or router.

Security Protocol Identifier

✓ This field from the outer IP header indicates whether the association is an AH or ESP security association.

Security Association Database

Security Parameter Index

- A 32-bit value selected by the receiving end of an SA to uniquely identify the SA. In an SAD entry for an outbound SA, the SPI is used to construct the packet's AH or ESP header.
- In an SAD entry for an inbound SA, the SPI is used to map traffic to the appropriate SA.

Sequence Number Counter

- A 32-bit value used to generate the Sequence Number field in AH or ESP headers.

Sequence Counter Overflow

- A flag indicating whether overflow of the Sequence Number Counter should generate an auditable event and prevent further transmission of packets on this SA.

Anti-Replay Window

- Used to determine whether an inbound AH or ESP packet is a replay.

AH Information

- Authentication algorithm, keys, key lifetimes, and related parameters being used with AH.

Security Association Database … Contd.

ESP Information

- Encryption and authentication algorithm, keys, initialization values, key lifetimes, and related parameters being used with ESP.

Lifetime of this Security Association

- A time interval or byte count after which an SA must be replaced with a new SA (and new SPI) or terminated, plus an indication of which of these actions should occur.

IPsec Protocol Mode

- Tunnel, transport, or wildcard.

Path MTU

- Any observed path maximum transmission unit (maximum size of a packet that can be transmitted without fragmentation) and aging variables.

Security Policy Database

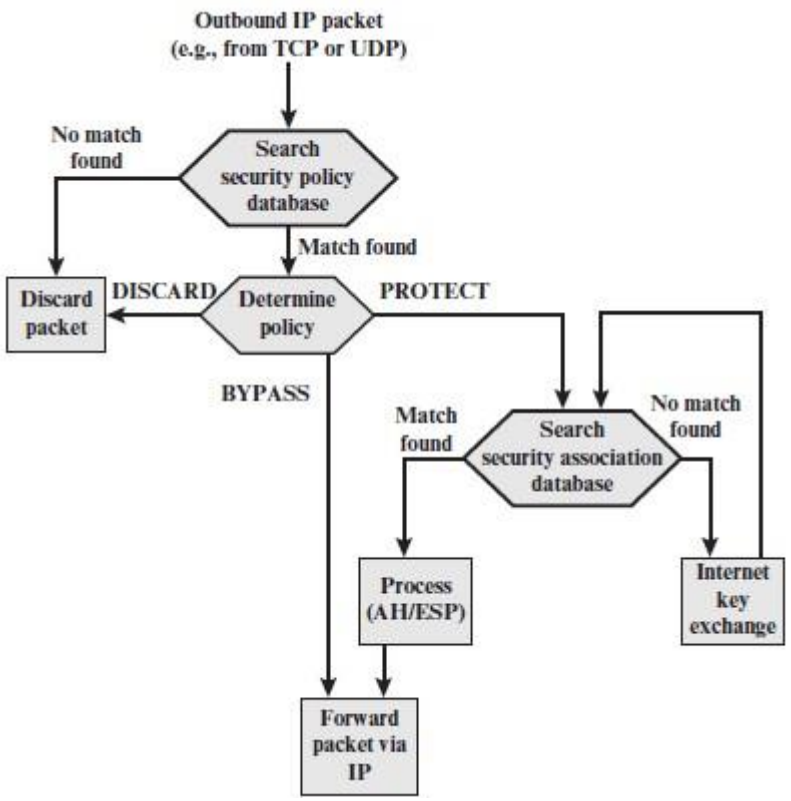
- relates IP traffic to specific SAs
  - match subset of IP traffic to relevant SA
  - use selectors to filter outgoing traffic to map
  - based on: local & remote IP addresses, next layer protocol, name, local & remote ports

Protocol	Local IP	Port	Remote IP	Port	Action	Comment
UDP	1.2.3.101	500	*	500	BYPASS	IKE
ICMP	1.2.3.101	*	*	*	BYPASS	Error messages
*	1.2.3.101	*	1.2.3.0/24	*	PROTECT: ESP intransport-mode	Encrypt intranet traffic
TCP	1.2.3.101	*	1.2.4.10	80	PROTECT: ESP intransport-mode	Encrypt to server
TCP	1.2.3.101	*	1.2.4.10	443	BYPASS	TLS: avoid double encryption
*	1.2.3.101	*	1.2.4.0/24	*	DISCARD	Others in DMZ
*	1.2.3.101	*	*	*	BYPASS	Internet

IP Traffic Processing

Outbound traffic

- A block of data from a higher layer, such as TCP, is passed down to the IP layer and an IP packet is formed, consisting of an IP header and an IP body.



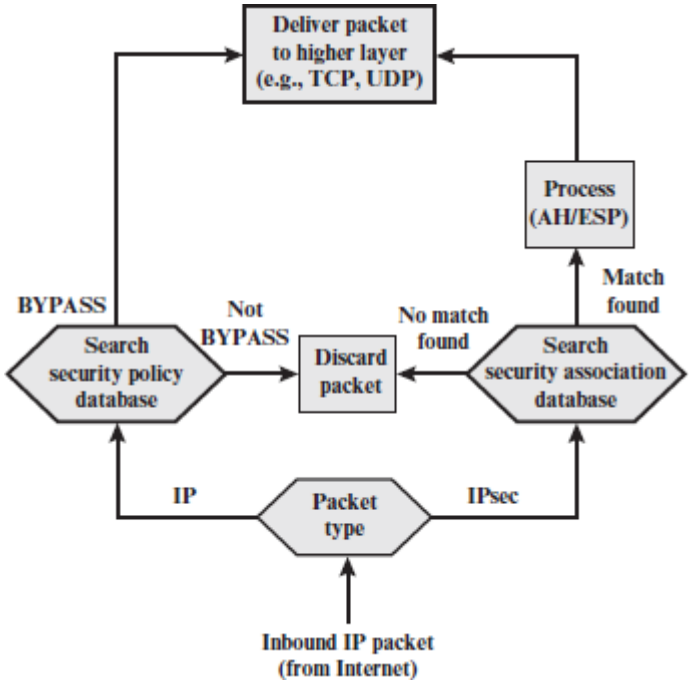
Steps

1. IPsec searches the SPD for a match to this packet.
2. If no match is found, then the packet is discarded and an error message is generated. If a match is found, further processing is determined by the first matching entry in the SPD. If the policy for this packet is DISCARD, then the packet is discarded. If the policy is BYPASS, then there is no further IPsec processing; the packet is forwarded to the network for transmission.
3. If the policy is PROTECT, then a search is made of the SAD for a matching entry. If no entry is found, then IKE is invoked to create an SA with the appropriate keys and an entry is made in the SA.
4. The matching entry in the SAD determines the processing for this packet. Either encryption, authentication, or both can be performed, and either transport or tunnel mode can be used. The packet is then forwarded to the network for transmission.

IP Traffic Processing … Contd.

Inbound traffic

- An incoming IP packet triggers the IPsec processing.



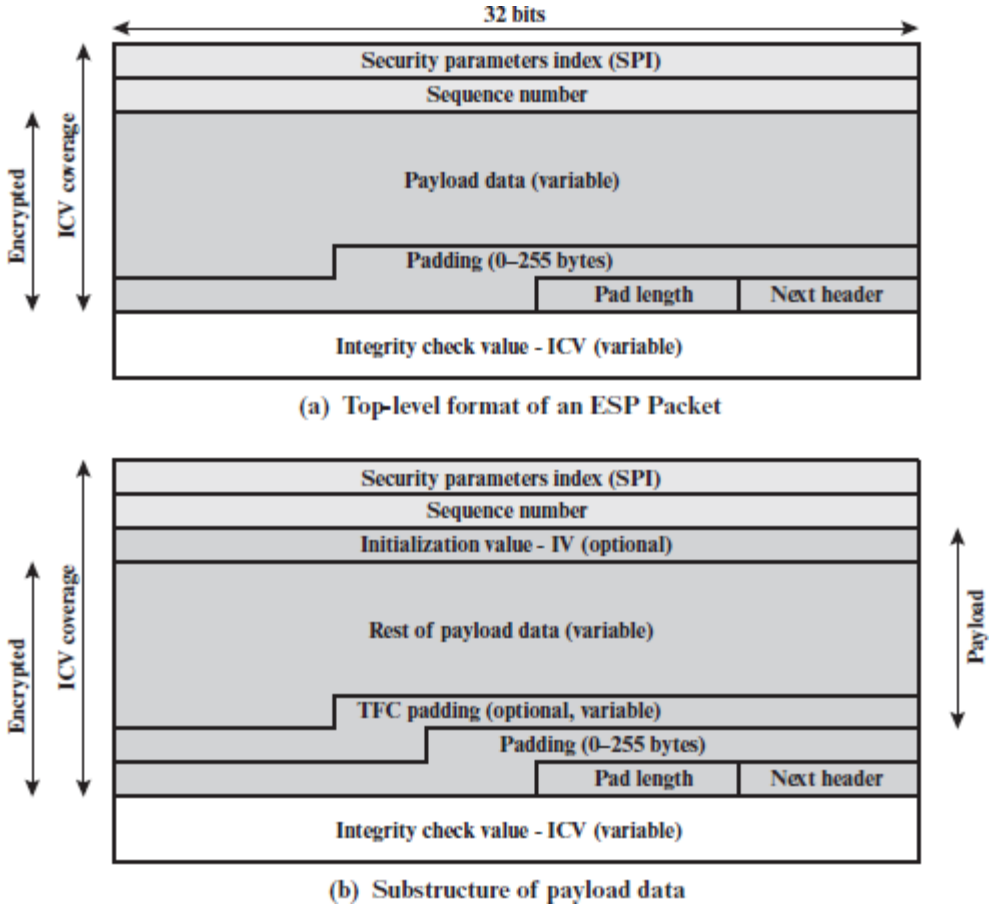
Steps

1. IPsec determines whether this is an unsecured IP packet or one that has ESP or AH headers/trailers, by examining the IP Protocol field (IPv4) or Next Header field (IPv6).
2. If the packet is unsecured, IPsec searches the SPD for a match to this packet. If the first matching entry has a policy of BYPASS, the IP header is processed and stripped off and the packet body is delivered to the next higher layer, such as TCP. If the first matching entry has a policy of PROTECT or DISCARD, or if there is no matching entry, the packet is discarded.
3. For a secured packet, IPsec searches the SAD. If no match is found, the packet is discarded. Otherwise, IPsec applies the appropriate ESP or AH processing. Then, the IP header is processed and stripped off and the packet body is delivered to the next higher layer, such as TCP.

Encapsulating Security Payload (ESP)

- provides message content confidentiality, data origin authentication, connectionless integrity, an anti-replay service, limited traffic flow confidentiality
- services depend on options selected when establish Security Association (SA), net location
- can use a variety of encryption & authentication algorithms

ESP Packet Format

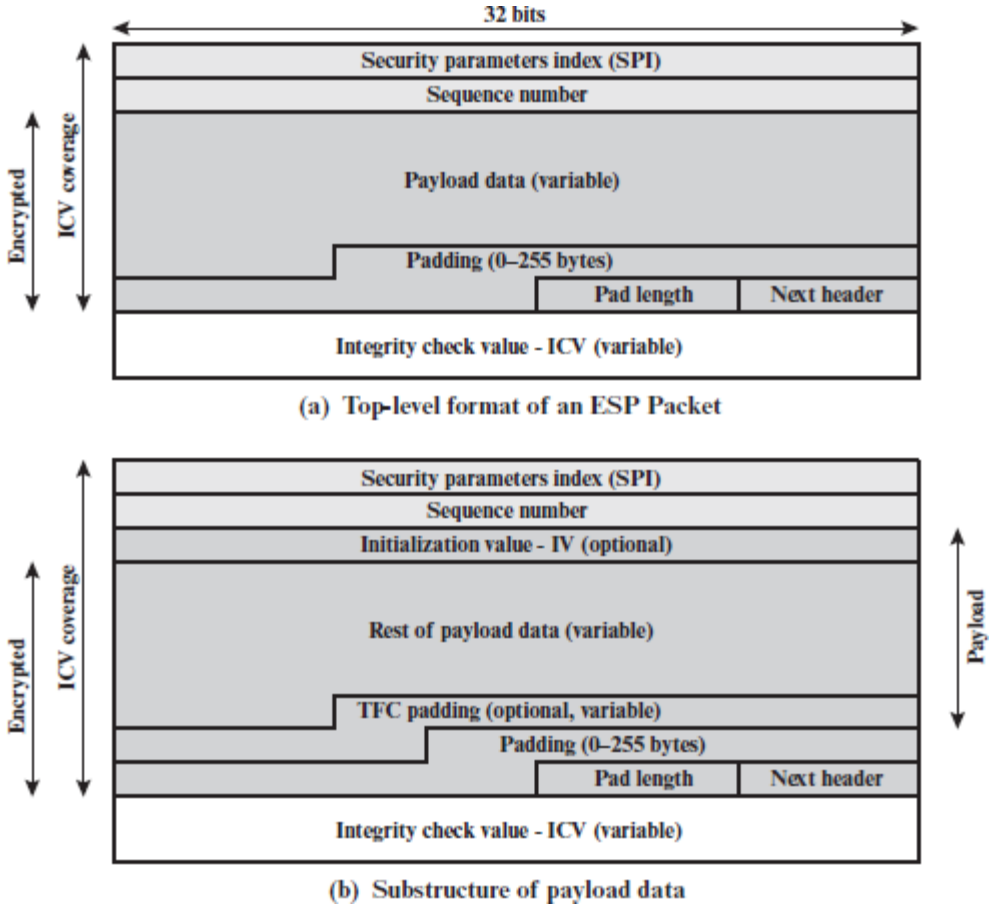


(a) Top-level format of an ESP Packet

(b) Substructure of payload data

- **Security Parameters Index (32 bits):** Identifies a security association.
- **Sequence Number (32 bits):** A monotonically increasing counter value; this provides an anti-replay function.
- **Payload Data (variable):** This is a transport-level segment (transport mode) or IP packet (tunnel mode) that is protected by encryption.
- **Padding (0–255 bytes):** Expanding plaintext.
- **Pad Length (8 bits):** Indicates the number of pad bytes immediately preceding this field.
- **Next Header (8 bits):** Identifies the type of data contained in the payload data field by identifying the first header in that payload (e.g., an extension header in IPv6, or an upper-layer protocol such as TCP).
- **Integrity Check Value (variable):** A variable-length field (must be an integral number of 32-bit words) that contains the Integrity Check Value computed over the ESP packet minus the Authentication Data field.

ESP Packet Format … Contd.



(a) Top-level format of an ESP Packet

(b) Substructure of payload data

- When any combined mode algorithm is employed, the algorithm itself is expected to return both decrypted plaintext and a pass/fail indication for the integrity check.
- For combined mode algorithms, the ICV that would normally appear at the end of the ESP packet (when integrity is selected) may be omitted.
- When the ICV is omitted and integrity is selected, it is the responsibility of the combined mode algorithm to encode within the Payload Data an ICV-equivalent means of verifying the integrity of the packet.
- Two additional fields may be present in the payload.
  - ✓ An **initialization value (IV)**, or nonce, is present if this is required by the encryption or authenticated encryption algorithm used for ESP.
  - ✓ If tunnel mode is being used, then the IPsec implementation may add **traffic flow confidentiality (TFC)** padding after the Payload Data and before the Padding field, as explained subsequently.

Encryption & Authentication

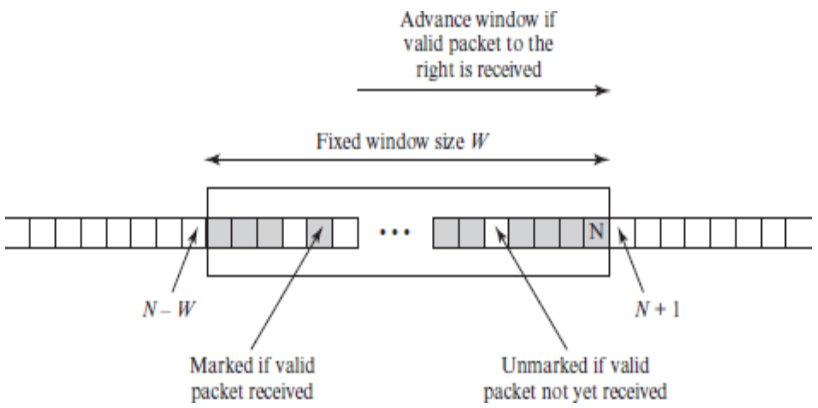
- ESP can encrypt payload data, padding, pad length, and next header fields
  - if needed have IV at start of payload data
- ESP can have optional ICV for integrity
  - is computed after encryption is performed

Padding

- ESP uses padding
  - to expand plaintext to required length
  - to align pad length and next header fields
  - to provide partial traffic flow confidentiality

Anti-Replay Service

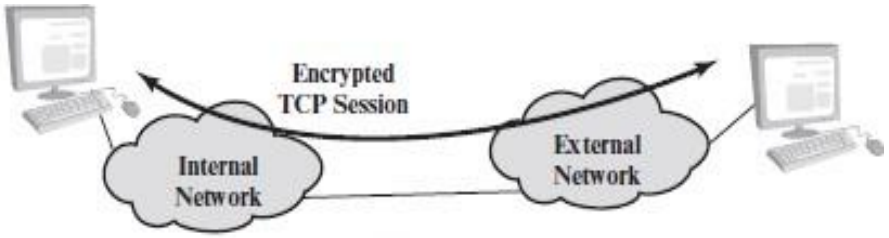
- replay is when attacker resends a copy of an authenticated packet
- use sequence number to thwart this attack
- sender initializes sequence number to 0 when a new SA is established
  - increment for each packet
  - must not exceed limit of $2^{32} - 1$
- receiver then accepts packets with seq no within window of $(N - W + 1)$



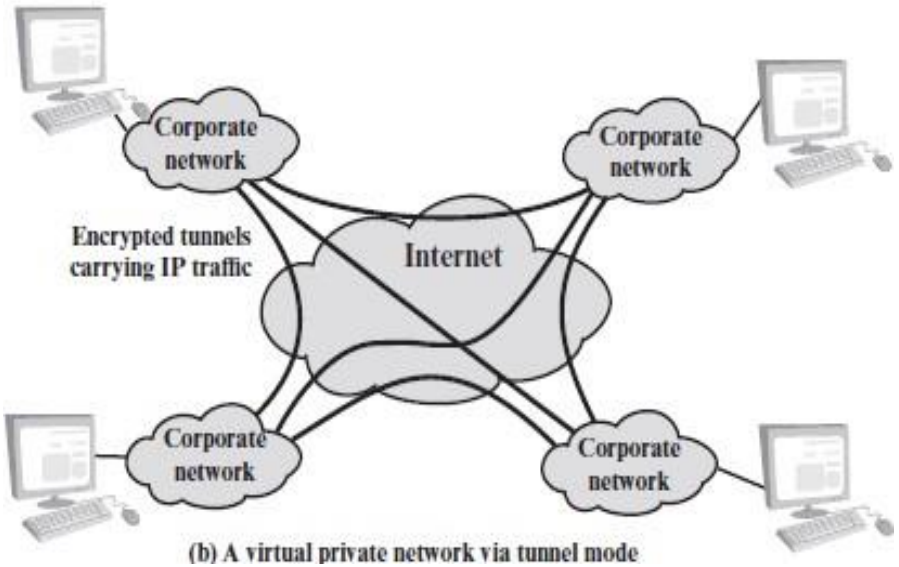
Inbound processing proceeds as follows when a packet is received:

1. If the received packet falls within the window and is new, the MAC is checked. If the packet is authenticated, the corresponding slot in the window is marked.
2. If the received packet is to the right of the window and is new, the MAC is checked. If the packet is authenticated, the window is advanced so that this sequence number is the right edge of the window, and the corresponding slot in the window is marked.
3. If the received packet is to the left of the window or if authentication fails, the packet is discarded; this is an auditable event.

Transport and Tunnel Modes



(a) Transport-level security

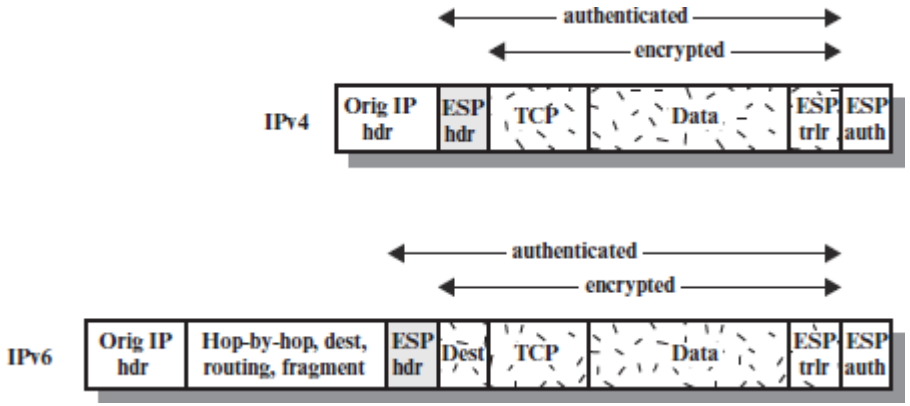


(b) A virtual private network via tunnel mode

- In Figure (a), encryption (and optionally authentication) is provided directly between two hosts.
- Figure (b) shows how tunnel mode operation can be used to set up a virtual private network.
  - ✓ An organization has four private networks interconnected across the Internet.
  - ✓ Hosts on the internal networks use the Internet for the transport of data but do not interact with other Internet-based hosts.
  - ✓ By terminating the tunnels at the security gateway to each internal network, the configuration allows the hosts to avoid implementing the security capability.

Transport and Tunnel Modes … Contd.

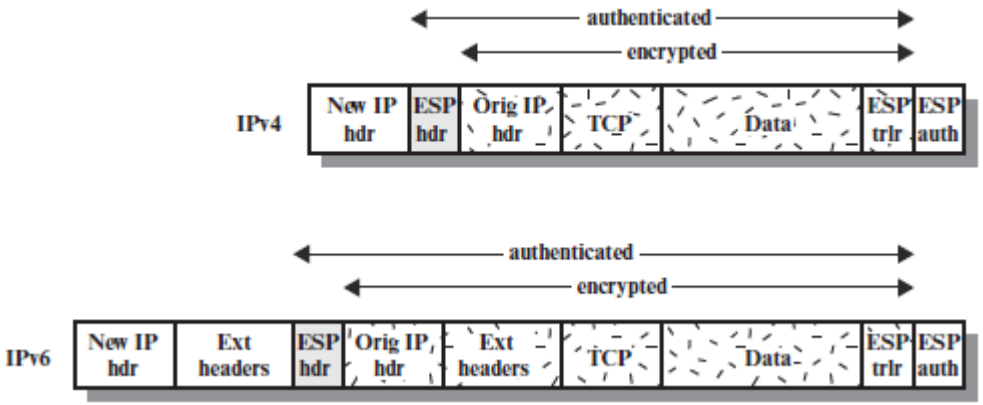
Transport Mode ESP



- At the source, the block of data consisting of the ESP trailer plus the entire transport-layer segment is encrypted and the plaintext of this block is replaced if this option is selected.
- The packet is then routed to the destination.
- Each intermediate router needs to examine and process the IP header plus any plaintext IP extension headers but does not need to examine the ciphertext.
- The destination node examines and processes the IP header plus any plaintext IP extension headers.
- On the basis of the SPI in the ESP header, the destination node decrypts the remainder of the packet to recover the plaintext transport-layer segment.
- Transport mode operation provides confidentiality for any application that uses it, thus avoiding the need to implement confidentiality in every individual application.
- One drawback to this mode is that it is possible to do traffic analysis on the transmitted packets.

Transport and Tunnel Modes … Contd.

Tunnel Mode ESP



- Tunnel mode ESP is used to encrypt an entire IP packet.
- For this mode, the ESP header is prefixed to the packet and then the packet plus the ESP trailer is encrypted.
- This method can be used to counter traffic analysis.
- Because the IP header contains the destination address and possibly source routing directives and hop-by-hop option information, it is not possible simply to transmit the encrypted IP packet prefixed by the ESP header.
- Intermediate routers would be unable to process such a packet.
- It is necessary to encapsulate the entire block (ESP header plus ciphertext plus Authentication Data) with a new IP header that will contain sufficient information for routing but not for traffic analysis.

Transport and Tunnel Modes … Contd.

Consider a case in which an external host wishes to communicate with a host on an internal network protected by a firewall, and in which ESP is implemented in the external host and the firewalls.

The following steps occur for transfer of a transport-layer segment from the external host to the internal host:

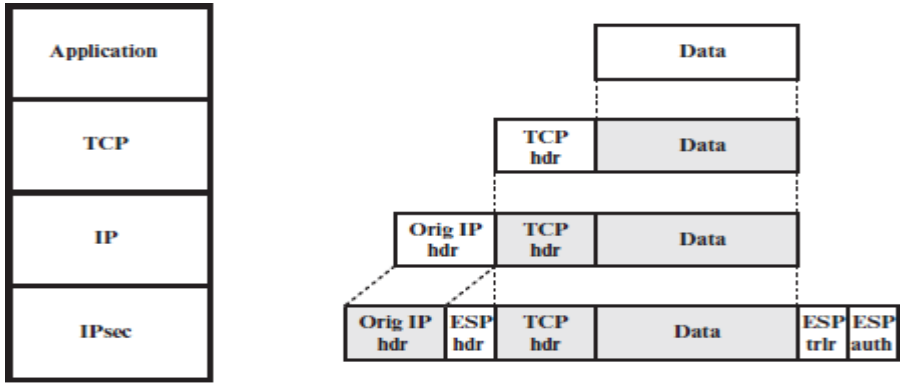
1. The source prepares an inner IP packet with a destination address of the target internal host.
   - ✓ This packet is prefixed by an ESP header; then the packet and ESP trailer are encrypted and Authentication Data may be added.
   - ✓ The resulting block is encapsulated with a new IP header (base header plus optional extensions such as routing and hop-by-hop options for IPv6) whose destination address is the firewall; this forms the outer IP packet.

1. The outer packet is routed to the destination firewall.
   - ✓ Each intermediate router needs to examine and process the outer IP header plus any outer IP extension headers but does not need to examine the ciphertext.

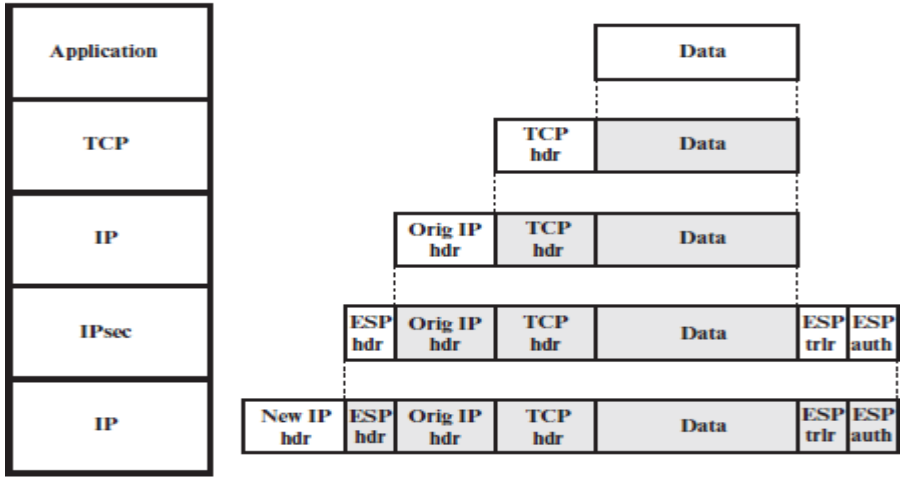
2. The destination firewall examines and processes the outer IP header plus any outer IP extension headers.
   - ✓ Then, on the basis of the SPI in the ESP header, the destination node decrypts the remainder of the packet to recover the plaintext inner IP packet.
   - ✓ This packet is then transmitted in the internal network.

3. The inner packet is routed through zero or more routers in the internal network to the destination host.

Protocol Architecture



(a) Transport mode



(b) Tunnel mode